MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, CMOS (INITIAL LOGIC)

This specification is approved for use by the U.S. ARMY Arament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

This specification contains requirements not covered by the drawings and provides quality assurance provisions for fabrication and packing of a complementary, metal oxide, semiconductor (CMOS) integrated circuit used in the MINE, antipersonnel: HE, M74 and MINE, antitank: HE, M75.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits , General Specification for
MIL-M-55565 - Microcircuits, Preparation for Delivery of

STANDARDS

MIL-STD-883 - Test Methods and Procedures for Microcircuits
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

FSC : 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DELOPMENT COMMAND (ARRADCOM)

9292962 - Microcircuit, Digital, CMOS (Initial Logic)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement [unctions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Device classitication. The microcircuit (device) shall conform to the product assurance level class B of MIL-M-38510 and as specified herein.

3.2 Device. The device shall comply with all requirements specified on Dwg. 9292962 and with all requirements specified in applicable specifications and standards.

3.3 Design and construction. The design and construction shall comply with the requirements of MIL-M-38510 for class B devices and 3.2 herein.

3.3.1 Glassivation. The microcircuit shall be glassivated with silicon dioxide to render it impervious to a contaminated environment. The glassivation shall cover all electrical conductors except the bonding pads.

3.4 Environmental and mechanical. Devices to be delivered under this specification shall be capable of meeting the following MIL-STD-883 environmental and mechanical tests when the devices are subjected to these tests in accordance with this specification.

Environmental tests	Method	Conditions
Moisture resistance	1004	
Steady state life	1005	D, (1000 hrs @ 125°C)
Temperature cycling	1010	C, 10 cycles
Seal (fine and gross)	1014	(A or B & C, respectively)
High temperature storage	1008	c
Burn in	1015	D, 168 hrs @ 125°C

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Environmental tests	Method	Conditions
Mechanical shock	2002	D
Solderability	2003	
Lead integrity	2004	B2
Resistance to solvents	2015	
Vibration, variable frequency	2007	A
External Visual	2009	
Internal Visual	2010	B
Constant acceleration	2001	E, Y_1, orientation only
Bond strength	2011	D
Physical dimensions	2016	

3.5 Electrical requirements. The device shall have the following electrical characteristics.

3.5.1 Operating ratings: See Table I.

TABLE 1, Operating Ratings.

CHARACTERISTIC	SYMBOL	RATING	UNIT
DC Supply Voltage	V_{DD}	-0.5 to 7.5	Volts
All Inputs	V_{IN}	$V_{IN} \leq V_{DD}$	Volts
Temperature			
Operating 1/	$T_{A(OP)}$	-45 to +85	°C
Storage	$T_{A(STG)}$	-55 to +125	°C

1/ Device shall be capable of burn-in at +125°C. (See 4.5.1.7)

3.5.2 Performance characteristic. The device shall meet the performance characteristics of Table II.

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TABLE II. ELECTRICAL PARAMETERS.

CHARACTERISTICS	PINS	V_{DD}	T_A 1/	MIN	MAX	UNIT	2/	3/
Input Current	1, 2, 10, 11, 13, 15	7.5 v			100	nA	3.5.2.1	4.5.4
Leakage Current Oscillator		7.5 v			35	uA	3.5.2.2	4.5.5
Leakage Current Low		7.5 v	+25°C		4	uA	3.5.2.3	4.5.6
		7.5 v			10	uA		
Input Current	1	5.0 v		0.75	5	uA	3.5.2.4	4.5.7
	15	5.0 v		0.50	5	uA	3.5.2.5	4.5.8
Output Low @ (1.0)v	3, 4, 5, 6	5.0 v		50		uA	3.5.2.6	4.5.9
	7, 9	5.0		1.25		mA	3.5.2.7	4.5.9
	12, 14	5.0		100		uA	3.5.2.8	4.5.9
Output High @ (V_{DD}-1.0)v	3, 4, 5 6, 7, 9	5.0		50		uA	3.5.2.9	4.5.10
	12, 14	5.0		100		uA	3.5.2.10	4.5.10
Switching point 4/	2	5.0		2.5	4.3	V	3.5.2.11	4.5.11
	13	7.5		2.5	5.0	V	3.5.2.12	4.5.12
Noise Immunity	1, 10, 11, 15	5.0		1.0		V	3.5.2.13	4.5.13
		7.5		1.5		V		
Functional Characteristics							See 3.5.3	4.5.14

1/ Unless otherwise specified, the MIN/MAX values apply over the temperature range -45°C $\leq T_A \leq$ +85°C.

2/ Requirement paragraph (assigned in this table).

3/ Test Method paragraph.

4/ Switching shall occur between applicable max and min voltage level.

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3.5.3 Functional operations. When the device is connected as shown in Figure 4 and the waveforms shown in Figures 5 thru 14 are applied to terminals 1, 2, 10, 11, 13 and 15; operation shall be as described by the output waveforms at terminals 3, 4, 5, 6, 7, and 9 in Figures 5 thru 14 when tested as specified in 4.15.14.

3.5.4 Oscillator function. (Pins 12, 13, 14). The device shall function as an oscillator with the following output periods at pin 5:

(a) When the temperature varies over the operating range (-45°C to 85°C) and the voltage varies from 3.5 to 4.5 volts, the output period shall not shift more than ±5% from the period at +25°C and V_{DD} = 4.0 volts.

(b) When the temperature varies over the operating range (-45°C to +85°C) and the voltage varies from 5 to 7.5 volts, the output period shall not shift more than ±5% from the period at 25°C and V_{DD}= 6.2 volts.

(c) The output period at pin 5 shall be 173 ±35 millisecond @ 25°C and V_{DD} = 7.0 volts.

These requirements shall be tested for as specified in 4.5.15.

3,5,5 Reset pins/bond pads. The reset line on the microcircuit shall be accessible to external signals via dual bond pads, with each bond pad connected to its own package pin (pins 10, 11). When tested as specified in 4.5.16, the maximum resistance between pins 10 and 11 shall be 10K OHMS.

3.5.6 Screening. All devices to be delivered in accordance with this specification shall have been subjected to the screening procedures as specified in MIL-STD-883 Method 5004 for a class B device, and as specified herein (see 4.4.2.1).

3.5.6.1 Burn-in. The devices shall be subjected to the burn-in test as specified in MIL-STD-883 Method 5004 and 4.5.17 herein.

3.5.7 Steady state life. The devices shall be subjected to the steady state life test of MLL-STD-883 Method 1005 condition D and 4.5.18 herein.

3.6 Device marking. The device shall be marked with the intoxmation specified in 3.6.1 thru 3.6.5, and the primary markings (index point, part number, and inspection lot identification code) shall be located on the top surface of the package. The other markings may be placed in any suitable location so as to perform their functions and not interfere with the primary markings.

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3.6.1 Index point. See 3.2.

3,6.2 Part number. The complete part number shall be as follows:

9292962X

where 9292962 is the microcircuit Drawing number and X the drawing revision letter, if applicable.

3.6.3 Inspection lot identification code. The inspection lot identification code marking shall be in accordance with MIL-M-38510.

7.6.4 Manufacturer's designating symbol. The manufacturer's designating symbol marking shall be in accordance with MIL-M-38510.

3.6.5 Other markings. The vendor may add additional markings, for his own use, on the bottom surface of the package provided that these markings do not interfere with the required markings.

3.7 Data requirement. Data shall be recorded and retained for all tests required by this specification in accordance with 4.5.20.

3.8 Process controls. All process controls essential for the manufacturing of microcircuits to assure a reliable end item shall he submitted to the designated agency specified in 6.5 herein, prior to the start of fabrication.

3.8.1 Metallization. At random intervals, the Government Inspector shall monitor the contractor's quality and acceptability of metallization process on the semiconductor devices.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. (See MIL-M-38510).

3.11 Country of origin. The device shall be manufactured and sold by a qualified producer in the United States or in another country which is a party to one of the existing international agreements ratified by the United States, such as the NATO Electronic Parts, and the United States/Canada/Australia Agreement for Qualification of Products for Non-resident Manufacturers.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-M-38510 shall apply and are hereby made part of this detail specification, The supplier is responsible for the performance of all inspection requirements as specified herein, and the supplier may use his own or any other facilities suitable for the performance of the tests and inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to witness or to perform any of these tests and inspections set forth in this specification where such test and inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The Government reserves the right to audit the data resulting from the manufacturer's performance of these tests and inspections.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

c. Screening

4.3 First article inspection,

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article inspection lot size after screening shall consist of fifty (50) devices, minimum.

4.3.2 Inspection to be performed. First article inspection shall be in accordance with the applicable requirements of MIL-M-38510 and as modified by this specification, except that the procuring agency shall be the qualifying activity. First article requirements shall consist of inspections specified in subgroups "a" through "g" listed in TAble 111. First article items may be subjected by the qualifying activity (Government) to all of the examinations and tests specified in this specification and to any requirements of the applicable drawings.

a. Screening. Screening shall be performed on all devices in accordance with the inspection flow specified in 4.4.2.1.

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4.3.3 Rejection. If any device fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Goverment reserves the right to terminate its inspection upon any failure of a device in the sample to comply with any of the stated requirements.

4.3.4 Requalification. The device shall be subjected to another First Article as stipulated in the contract if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the device are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

h. The contractor relocated his assembly facilities.

c* The contractor uses a new sub-contractor for material, parts, or assembly.

(-i. A change occurs in devices's material, performance or assembly.

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS (Initial logic)				9292962
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Screening	1/		3.5.6	4.4.2.1
	Subgroup a	25			
	Electrical parameters @ -45°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Electrical parameters @ +85°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Oscillator function			3.5.4	4.5.15
	Reset pins/bond pads			3.5.5	4.5.16
	Subgroup b	25			
	Mechanical shock			3.4	MIL-STD-883, Method 2002, D
	Vibration, variable frequency			3.4	MIL-STD-883, Method 2007, A
	Moisture resistance			3.4	MIL-STD-883, Method 1004
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, A, or B
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014, C
	External visual			3.3,3.4, 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Country of origin			3.11	Visual
	Subgroup c	25			
	Resistance to solvents	3		3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls 2/)			3.4	MIL-STD-883, Method 2011, D
	Subgroup d	25			
	Solderability (leads inspected 3/)			3.4	MIL-STD-883, Method 2003

NOTES: See page 2. FIRST ARTICLE Sample of 50 Devices: 25 for groups a-f; 25 for group c.

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-[illegible]A AR

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS (Initial logic)	2 OF 3	9292962
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
	Subgroup e				
	Static charge protection circuit test	6		3.2	4.5.19 herein
	Subgroup f	2			
	Lead integrity			3.4	MIL-STD-883, Method 2004,
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, or B
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014,
	Subgroup g	25			
	Steady state life			3.5.7	4.5.18
	Electrical parameters @ 25°C ± 5°C			3.5.2	Table II, 4.4.3.1

Notes:

1/ Devices which fail any test criteria in the screening sequence shall be removed from the First article inspection lot. The contractor should choose a First Article inspection lot size that is adequate for the needs of the First Article Inspection.

2/ The sample size is the number of bond pulls selected from a minimum number of 10 devices.

3/ The sample size is the number of leads inspected: In no case shall less than 3 devices be used to provide the number of required leads.

4/ All electrical parameters shall be read and recorded as go/no go data.

NOTES:

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TABLE III. FIRST ARTICLE INSPECTION.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63320A(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9292962
	MICROCIRCUIT, DIGITAL, CMOS (Initial logic)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	5/ The same sample units shall be subjected to all the tests in a subgroup. 6/ Devices used in subgroup a shall be used in subgroup b. Devices used in subgroups a and b may be used in subgroups c, d, e and f. 7/ Tests within each subgroup shall be performed in the sequence shown.				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lot fomation shall be as defined in MIL-M-38510 for Class B devices.

4.4.2 Examinations,

a. Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified for Group A and Croup B and when required Group C. Quality conformance inspection requirements shall be in accordance with the Classification of Defects and Test paragraph 4.4.2.1 through 4.4.2.4.

h. Inspection requirements. Inspection requirements for this device shall he in accordance with the requirements of MIL-M-38510 and MTL-STD-883, Method 5005 for Groups A, B. and C tests except as modified herein. See 4.4.2.1 thru 4.4.2.4.

c. Screening shall be performed on all devices in accordance with the inspection flow specified in 4.4.2.1.

d. Sampling. Inspection samples of devices required by this specification shall be randomly selected from each inspection lot in accordance with Appendix B-1 of MIL-M-38510

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Microcircuit, Digital, CMOS (INITIAL LOGIC)				9292962
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	SCREENING: (3.4, 3.5.6)				
	Internal visual (Pre-cap) 1/		100%	3.5.6 3.4,3.5.1	MIL-STD-883, Method 2010, B
	High temperature storage (post seal stabilization bake)		100%	3.5.6	MIL-STD-883, Method 1008, C
	Temperature cycling		100%	3.5.6	MIL-STD-883, Method 1010, C
	Constant acceleration		100%	3.5.6	MIL-STD-883, Method 2001, E, Y1 orientation only
	Seal (fine leak)		100%	3.5.6	MIL-STD-883, Method 1014, A or B
	Seal (gross leak)		100%	3.5.6	MIL-STD-883, Method 1014, C
	Electrical parameters @ 25°C (pre burn-in)		100%	3.5.6	Table II, 4.4.3.1 herein
	Burn-in		100%	3.5.6.1	4.5.17 herein
	Electrical parameters @ 25°C (post burn-in)		100%	3.5.2	Table II, 4.4.3.1 herein
	External visual		100%	3.5.6, 3.3,3.4, 3.5.6,3.6 3.10	MIL-STD-883, Method 2009
	Country of origin		100%	3.11	Visual

NOTES: 1/ See 3.8.1: Also random monitoring of metallization process by Government inspector.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-38320A/AR

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Microcircuits, Digital, CMOS (INITIAL LOGIC)				9292962
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST GROUP A: (see 4.4.3.2, 4.4.3.3)	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electrical parameters @ -45°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Electrical parameters @ +85°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Oscillator function		LTPD 15%	3.5.4	4.5.15
	Reset pins/bond pads		LTPD 15%	3.5.5	4.5.16

NOTES:

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63320A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Microcircuit, Digital, CMOS (INITIAL LOGIC)				9292962
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST GROUP B: (see 4.4.3.4)	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup 1				
	Physical dimensions	4	0-1	3.2,3.4	MIL-STD-883, Method 2016
	Subgroup 2				
	Resistance to solvents	3	0-1	3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls 1/)		LTPD 15	3.4	MIL-STD-883, Method 2011, D
	Subgroup 3				
	Solderability (leads inspected 2/)		LTPD 15	3.4	MIL-STD-883, Method 2003

NOTES: 1/ The LTPD sample size is the number of bond pulls selected from a minimum number of 10 devices. 2/ The LTPD sample size is the number of leads inspected: In no case shall less than 3 devices be used to provide the number of leads required.

DD Form 1 JUL 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Microcircuit, Digital, CMOS (INITIAL LOGIC)				9292952
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST GROUP C: (see 4.4.3.5)	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
	Subgroup 1				
	Steady state life		LTPD 5%	3.5.7	4.5.18
	Electrical parameters @ 25°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Subgroup 2				
	Mechanical shock		LTPD 15%	3.4	MIL-STD-883, Method 2002,
	Vibration, variable frequency		LTPD 15%	3.4	MIL-STD-883, Method 2007,
	Seal (fine leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014, or B
	Seal (gross leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014,
	External Visual		LTPD 15%	3.3,3.4 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C		LTPD 15%	3.5.2	Table II, 4.4.3.1 herein
	Packaging		2/	5.1	MIL-M-55565
	Packing		2/	5.1	MIL-M-55565
	Marking		2/	5.2	MIL-M-55565 & MIL-STD-118

NOTES: 1/ The same sample units shall be subjected to all the tests in a subgroup.
2/ Sampling and inspection as per MIL-M-55565.

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection "Classification of Defects and Tests" forms and below.

4.4.3.l Electrical parameters @ 25° ± 5°C testing. Each device shall be tested for compliance with the requirements listed in Table II. Any device classified as defective shall be removed from the lot.

4.4.3.2 Testing sequence. The order of group testing shall be A, B, and C: Only lots that passed Group A tests shall be submitted for Groupd B testing, and only lots that passed Group B tests shall be submitted for Group C testing.

4.4.3.3 Croup A testing. Random samples from each inspection lot shall be selected, in accordance with this specification, and tested at the minimum and maximum temperatures for compliance with the requirements listed in Table II. An inspection lot which fails the Group A test shall be classified as defective until all the defective devices within the lot are identified and removed by subjecting all the devices in the lot to the Group A test.

4.4.3.4 Group B testing. Random samples from each inspection lot that passed Group A tests shall be selected and tested for compliance with the requirements of the Group B tests in accordance with this specification. An inspection lot shall not be shipped if the devices of any subgroup of the Group B test fail to comply with all of the specified requirements.

4.4.3.5 Group C testing. Group C tests shall be performed on a continuing schedule of at least once every ninety (90) days while the device is being produced. Random samples from one inspection lot that passed Group B tests shall be selected and tested for compliance with the requirements of the Group C tests in accordance with this specifications. Further inspection lots shall not be shipped if the devices of any subgroup of the Group C tests fail to comply with all of the specified requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the Classification of Defects and Tests Forms and in 4.5 herein. The contractor shall submit for approval inspection equipment designs, including alternate equivalent design which he may desire to use, in accordance with the terms of the contract. See MIL-M-38510 and 6.3 herein.

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4.5 Test methods, circuits and procedures.

4.5.1 Alternate methods, circuits and procedures. Unless otherwise stated in the specific test method, the methods and circuits shown are given as basic measurement method. For alternate methods, circuits and procedure the manufacturer shall demonstrate to the procuring activity that alternates which he may desire and propose to use are equivalent and give results within the desired accuracy of measurement (see 6.3).

4.5.2 Defective. Any device failing to meet the applicable requirements in the following tests shall be classed and labeled defective and segregated from the inspection lot.

4.5.3 Test conditions. Unless otherwise specified all examinations and tests shall be made under the following conditions.

Temperature	25°C
Humidity	20 to 60 percent RH
Pressure	29 ±2 inches Hg
Voltage&Currents	±2%
Temperature (ALL) Tolerance	±5°C

4.5.4 Input current. Connect pins 10, 11, 13 and 15 to V_{DD} and measure the current into each pin, connect pins 1, 2, 10, 11 and 13 to V_{SS} (ground) and measure the current out of each pin. Any device failing to meet the max current requirement of 3.5.2.1 shall be classed defective. (Non-destructive test).

4.5.5 Leakage current oscillator. Measure the leakage current with the device connected as shown in Figure 1. Any device failing to meet the max current requirement of 3.5.2.2 shall be classed defective. (Non-destructive test).

4,5,6 Leakage current low. Measure the leakage current in accordance with the test sequence of Table IV and with the device connected as shown in Figure 2. Any device failing to meet the max current requirement of 3.5.2.3 shall be classed defective. (Non-destructive test).

4.5.7 Input current. Measure the current into pin 1 with pin 1 held at V_{DD}. Any device failing to meet the max and pin current requirement of 3,5.2.4 shall be classed defective. (Non destructive test).

4.5.8 Input current. Measure the current out of pin 15 with pin 15 held at V_{SS} (ground). Any device failing to meet the max and min current requirement of 3.5.2.5 shall be classed defective. (Non-destructive test).

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4.5.9 Output low. When the outputs at pins (3, 4, 5, 6, 7, 9, 12, 14) are sinking current and held at 1.0 volt, measure the current into each pin. Any device failing to meet the min current requirement of 3.5.2.6/3.5.2,7/3. 5.2.8 shall be classed defective. (Non-destructive test).

4.5,10 Output high. When the outputs at pins (3, 4, 5, 6, 7, 9, 12 and 14) are sourcing currents and held at (V_{DD} - 1.0) volts, measure the current out of each pin. Any device failing to meet the min current requirement of 3.5.2.9/3.5.2.10 shall be classed defective. (Non-destructive test).

4.5.11 Switching points . (Pin 2). Measure the switching points at pin 2 in accordance with the test sequence of Table V and with the device connected as shown in Figure 3. Any device failing to meet the switching requirement between the max and min voltage levels of 3.5.2.11 shall be classed defective. (Non-destructive test).

4.5.12 Switching point. (Pin 13). With pin 16 connected to 7.5 volts and pin 8 connected to V_{SS} (ground), set pin 13 at 2.5 volts and confirm that the output voltage @ Pin 12 is less than 1.5 volts. Set pin 13 to 5.0 volts and confim that the output voltage at pin 12 is greater than 6 volts. Any device failing to meet the min and max voltage requirements of 3.5.2.12 shall be classed defective, (Non- destructive test).

4.5.13 Noise immunity. Noise immunity at input pins (1, 10, 11, 15) shall be confirmed during the functional tests (see 4.5.14). Any device failing to meet the min noise voltage requirements of 3.5.2.13 for a V_{DD} of 5.0/7.5 volts shall be classed defective. (Non-destructive test).

4.5.14 Functional tests. The functional tests consist of ten sequential tests defined in Figure 5 thru 14. Tests defined by Figure 1 and 2 shall be run at V_{DD} = 5.0 volts and 7.5 volts; all other tests are to be run at 5.0 volts. At V_{DD} 5.0, input low and high logic levels shall be 1.0 volt and 4.0 volts respectively; output low logic level shall be less than 1.0 volts and output high logic level greater than 4.0 volts. At V_{DD} 7.5 volts, input low and high logic levels shall be 1.5 volts and 5.0 volts respectively; output low and high logic levels shall be less than 1.5 volts and greater than 5.0 volts respectively. Any device failing to meet the requirements of 3.5.3 shall be classed defective. (Non-destructive test).

4.5.15 Oscillator function test. The device shall be connected as shown in Figure 1 and tested for the requirements as specified in 3.5.4. Any device failing to meet the requirement of 3.5.4 shall be classed defective. (Non-destructive test).

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4.5.16 Reset pins/bond pads. The resistance specified in 3.5.5 shall be measured using a Government approved ohrmmeter. During this test, 5.0 ± 0.1 volts D.C. shall be applied between pins 8 and 16, and the drive current from the ohmmeter shall be 100 microampere maximum. Any device failing to meet the resistance requirement of 3.5.5 shall be classed defective. (Non-destructive test)..

4.5.17 Burn-in screen. All devices shall be subjected to the burn-in screen in accordance with MIL-STD-883, Method 1015, Condition D (168 hours minimum @ 125°C ± 5°C) and as specified herein. For this test the device shall be connected as shown in Figure 15 and conditioned at the specified test temperature. The power supply voltage applied to pin 16 shall be 7.5 volts dc and the conditioning signal applied to the input pin 13 shall be as shown in Figure 15 Any device failing to meet the requirement of 3.5.6.1 shall be classed defective. (Non-destructive test).

4.5.18 Steady state life. Devices shall be subjected to the steady state life test in accordance with MIL-STD-883, Method 1005, Condition D and as specified herein. Test duration shall be 100 hours minimum at 125°C ± 5°C. For this test the device shall be connected as shown in Figure 15, the same test set up as for the burn-in test. Electrical parameters shall be measured before, at 360 (+160, -0) hours into, and after the steady state life test in accordance with Method 1005 and Table II. Any device failing to meet the electrical parameters requirements of Table II shall be classed defective. (Non-destructive test).

4.5.19 Static discharge (V_{Zap}) test. All input device pins having input protection circuitry for static discharge shall be tested in accordance with the procedure specified in Figure 16.

4.5.20 Data requirements. The contractor shall submit lot traveler(s) and a quality inspection test! demonstration and evaluation report giving all detailed inspection and test results in accordance with Data Item Description DI-R-1724 (tailored) format to the technical agency listed in 6.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Devices to be delivered under this specification shall be prepared for delivery in accordance with the requirements of MIL-M-38510, Level C. Electro-static discharge (ES) protection in accordance with the requirements of MIL-M-55565 shall be provided,

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5.2 Marking Marking shall be in accordance with MIL-STD-1188.

5.3. Shipping. When microcircuit circuit packages of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The device covered by this specification is i ntended for use in U.S. Army Ordnance.

6.2 Ordering data. The procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Provisions for submission of first article samples.
(c) Quantity required and delivery schedules.
(d) Requirement for certification of conformance for each lot or shipment of product.

6.3 Submission of inspection equipment designs for approval. (See 4.4.4). Submit equipment designs as required to:

Commander
ARRADCOM
ATTN: DRDAR-QAA-I
Dover, NJ 07801

6.4 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Distribution of shop traveler inspection and test results, process controls. Shop traveler(s), inspection and test results (3,7,4.5.20), and process controls (3.8) shall be sent to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-M, Dover, NJ 07801.

6.6 Contract data requirements,

a. Contract data requirements for inspection equipment designs shall confom to data item description DI-E-1118 or DI-I-1714 with the appropriate addendum to each.

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b. Shop traveler(s) shall be submitted.

Process control documents shall be submitted in accordance with DI-P-1604 (Mod) with attachment 1 listed on DD Form 1423 incorporated in the contract.

d. Contract data requirements for quality inspection test report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results.

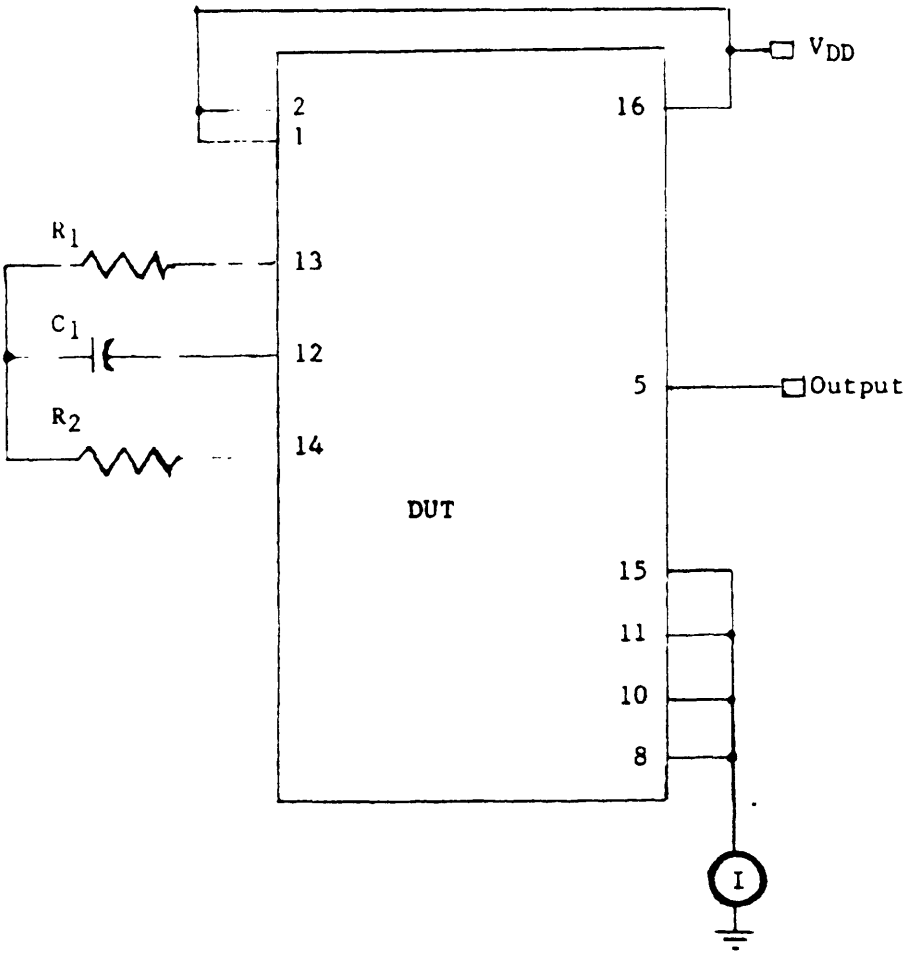
6.7 Changes from previous issue. Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

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Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A-151

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	NOMINAL	TOLERANCE	STYLE	TYPE	TCC
C1	0.01 μf	±1.0%	Ceramic	NPO	±25 ppm
R1	2.0 M	±5.0%	Composition		
R2	470 K	±1.0%	Metal Film		±50 ppm

FIGURE 1. Test Circuit for Leakage Current (Oscillator) and Oscillator Function tests.

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TABLE IV LEAKAGE CURRENT LOW TEST REQUIREMENTS.

Test Sequence No.	Test Condition: Figure 2 & Below
1	Set switch S_A to position "0" and S_B to position "1" to reset all the counters stages.
2	Measure leakage current.
3	Set switch S_B to position "0" and S_A to position "1" and load in 21,845 pulses at pin 13 to put the counters in a checker board pattern. Set switch S_A to position "0".
4	Measure leakage current.
5	Set switch S_A to position "1" and load in an additional 21,845 pulses (total 43,690) to put the counters in a reverse-checker board pattern. Set switch S_A to position "0".
6	Measure leakage current.
7	Set switch S_A to position "1" and load in an additional 21,845 pulses (total 65,535) to set all counter stages. Set switch S_A to position "0".
8	Measure leakage current.

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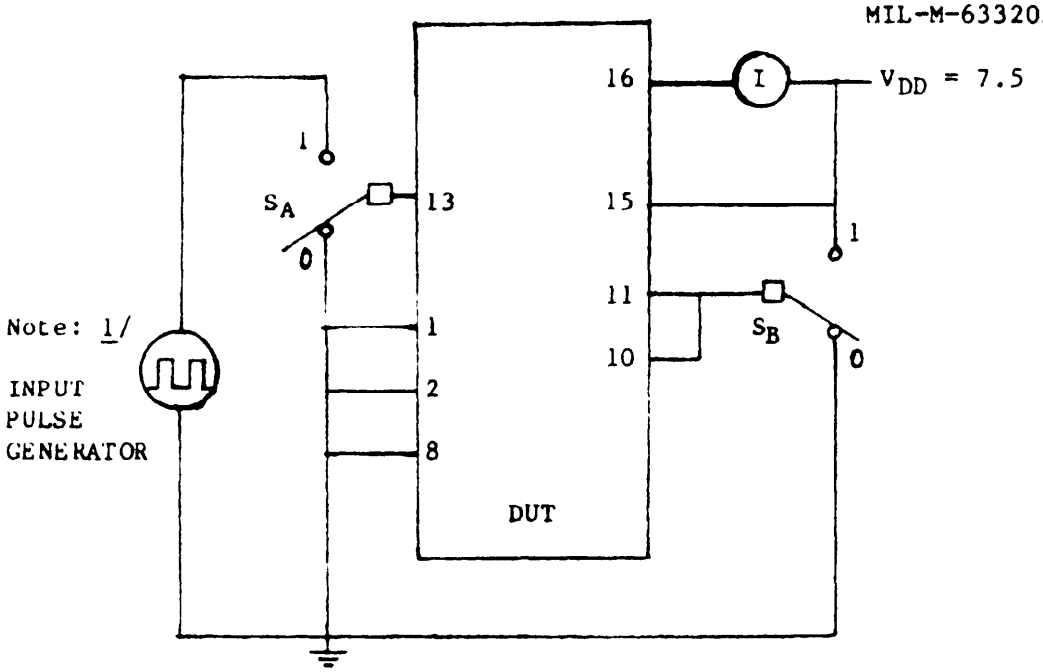


FIGURE 2. Test Set-up For Leakage Current Low Requirement.

1/ See Note 1 of Figure 4 for input pulse characteristics.

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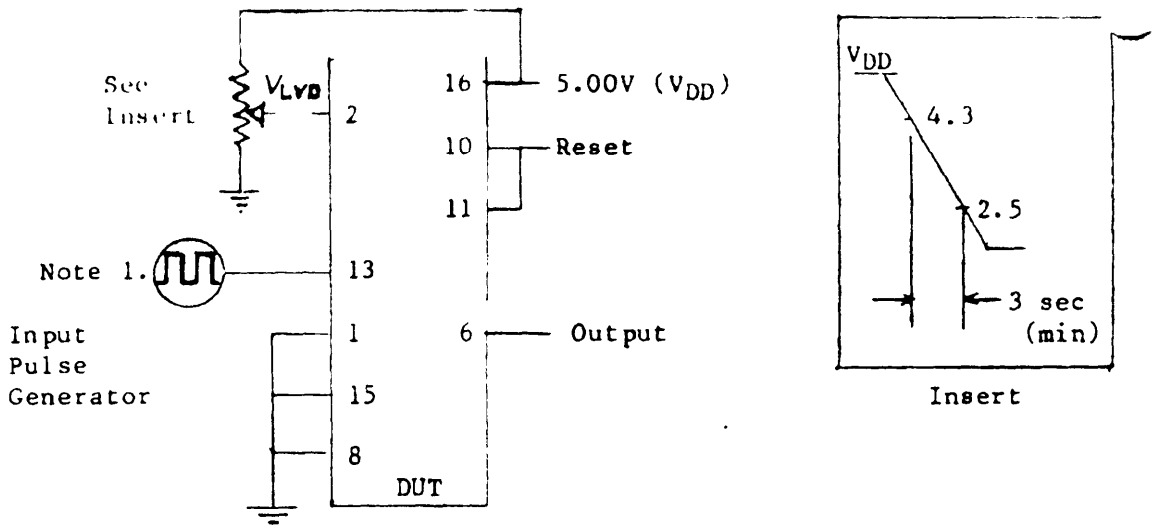


FIGURE 3. Test Setup for LVD Switching Point Requirment.

Note 1: See Figure 4 notes for input pulse characteristics.

TABLE V LVD Test Sequence.

Test Sequence	Test Conditions
1	Set pin 2 to V_{DD}, then reset and load in 8840 pulses at pin 13. Conform pin 6 is at logic "1" level.
2	Decrease the voltage at pin 2 to 4.3 volts and confirm pin 6 is at logic "1" level.
3	Linearly decrease the voltage at pin 2 from 4.3 to 2.5 volts and confirm pin 6 is at logic "0" level with pin 2 @ 2.5 volts.
	For First Article Tests measure and record switching point voltage at pin 2.

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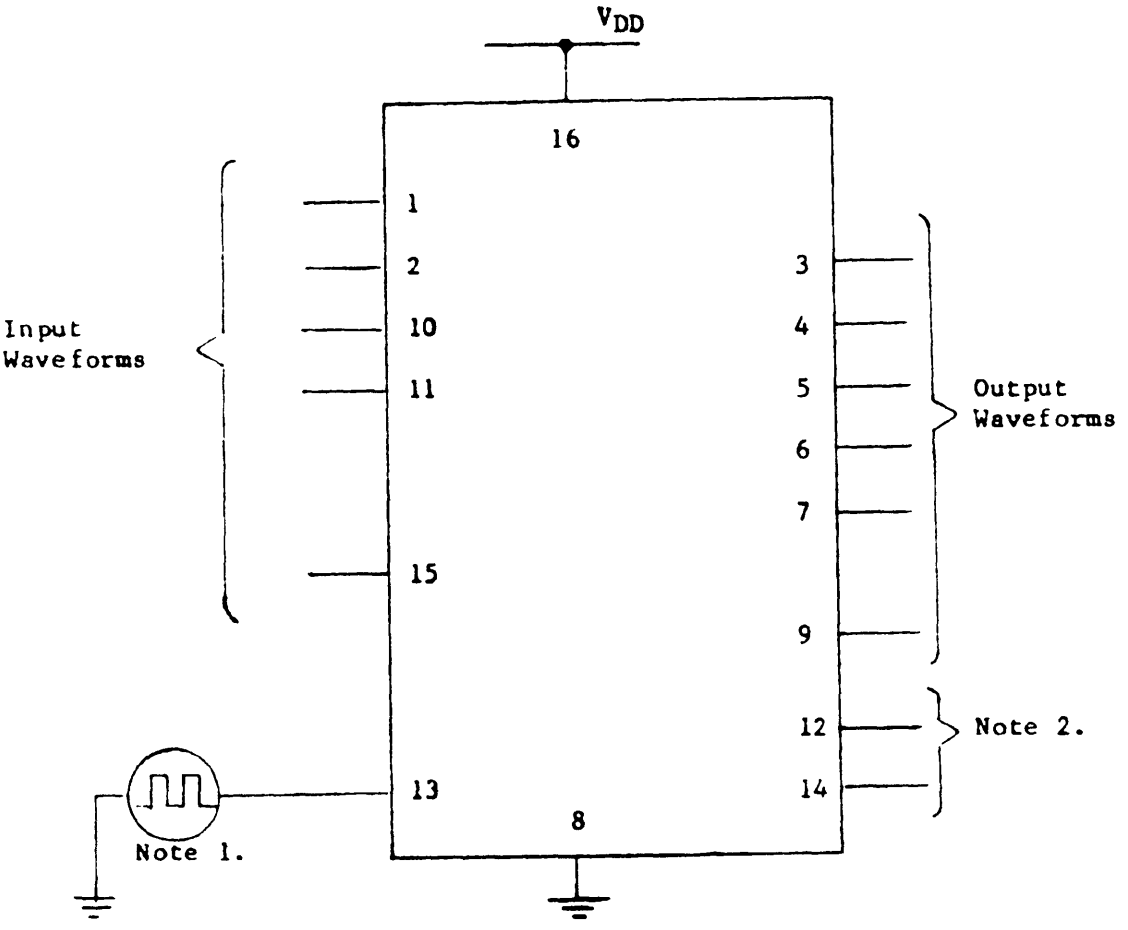


FIGURE 4. Test Setup For Functional Tests.

Note 1: Input Pulses shall have the following characteristics:

- Rise Time (t_R) = 1.0 microseconds max.
- Fall Time (t_f) = 1.0 microseconds max.
- Voltage (low) = 1.5 volts max @ V_{DD} = 7.5v
  1.0 volts max @ V_{DD} = 5.0v
- Voltage (high) = (V_{DD} - 1.5) volts min @ V_{DD} = 7.5v
  = (V_{DD} - 1.0) volts min @ V_{DD} = 5.0v
- Time high = 10.0 microseconds min.

Note 2: Outputs at pins 12 and 14 follow the input at pin 13, except that the output at pin 14 is inverted. The pulses at pin 14 will henceforth be identified as the primary clock input pulses and serve as the reference for pulse counts for this specification.

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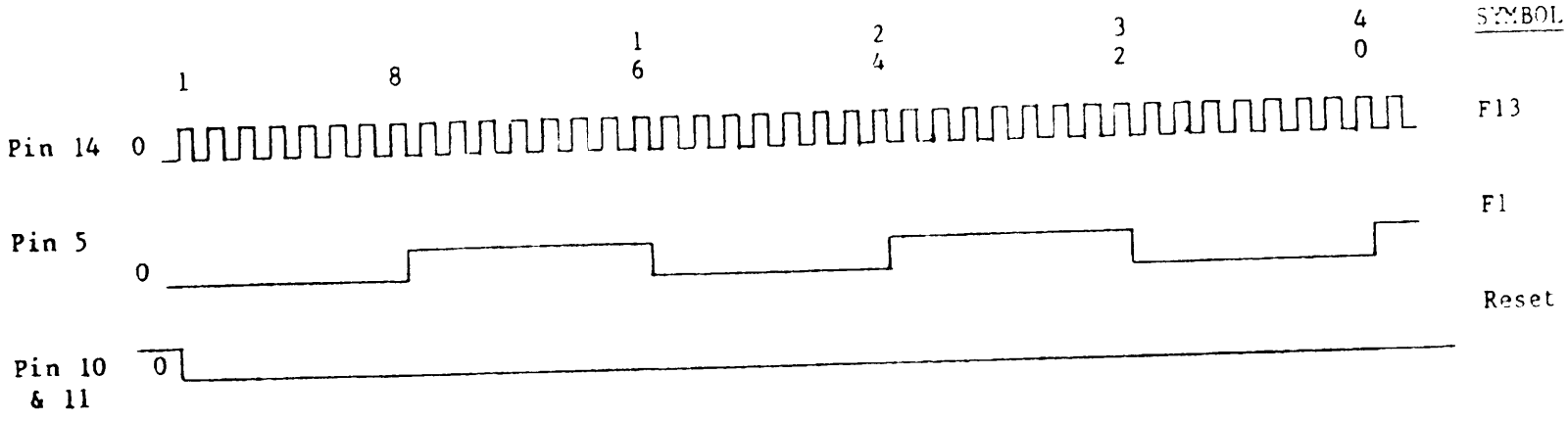


FIGURE 5. Functional Logic Timing Diagram (Pre-Scaler).

Functional logic relationship between primary clock input pulses at pin 14 and the output at pin 5.

Notes: Pins 1 and 15 shall be set at logic "0", pin 2 set at logic "1", pin 16 set to V_{DD}, pin 8 set to V_{SS} (ground); outputs at pins 3, 4, 7 and 9 shall be as defined in Figure 6.

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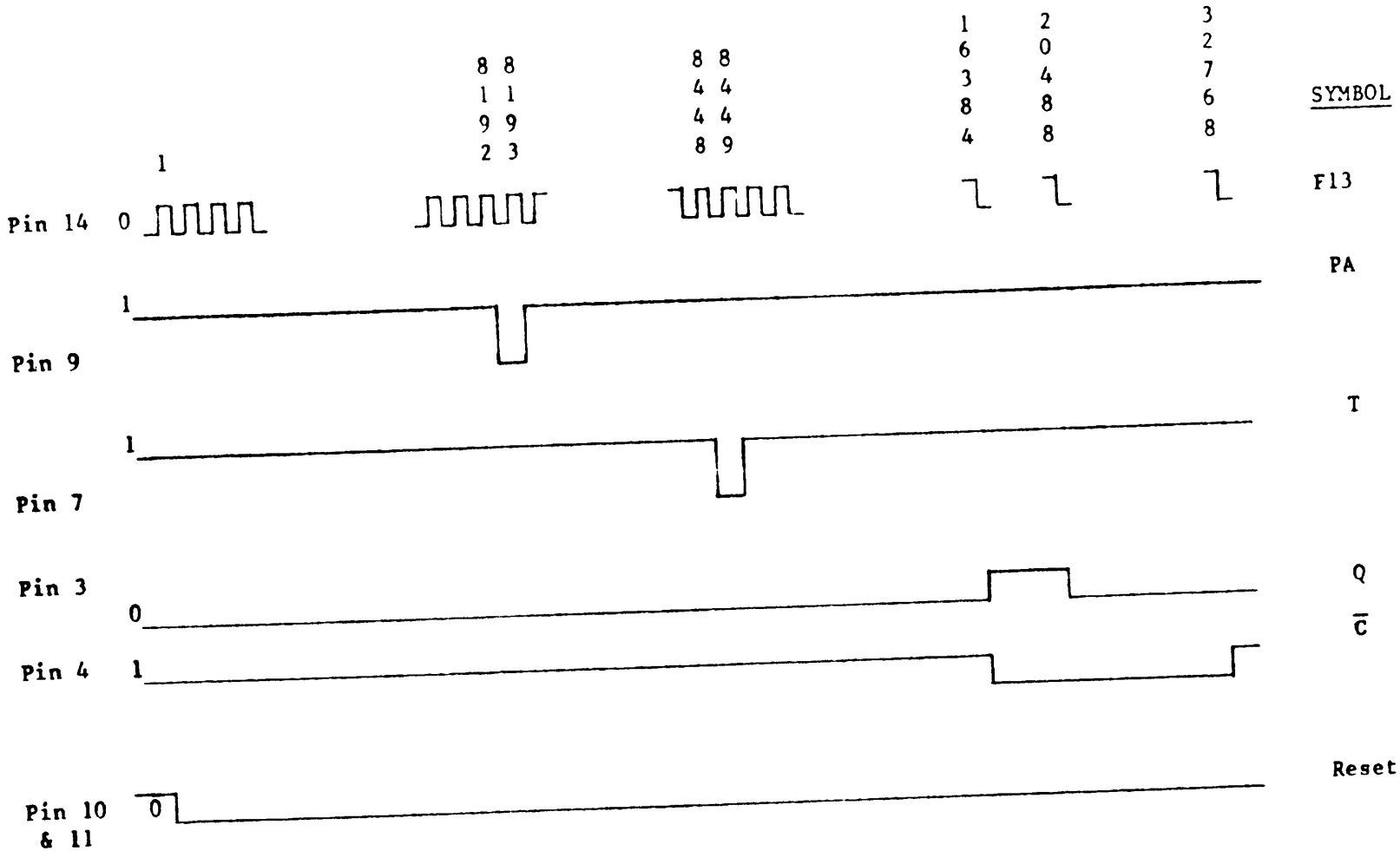


FIGURE 6. Functional Logic Timing Diagram (Output Signals PA, T, Q, C)

Functional logic relationship between primary clock input pulses at pin 14 and the outputs at pins 3, 4, 7, and 9.

Notes: Pins 1 and 15 shall be set at logic "0", pin 2 set at logic "1", pin 16 to V_{DD}, pin 8 to V_{SS} (ground), output pin 6 shall be at a logic "1" level, the output at pin 5 shall be as defined in Figure 5.

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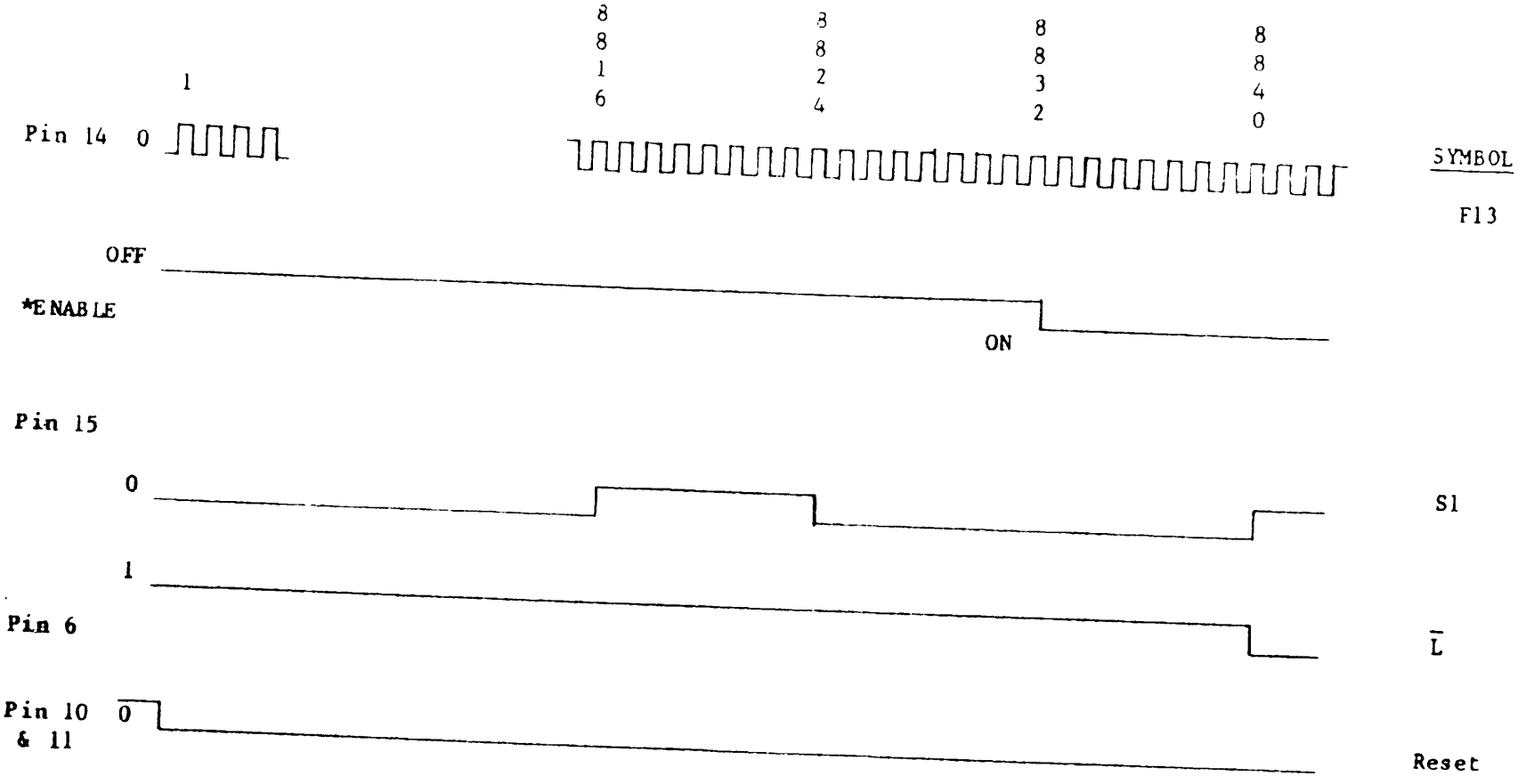


FIGURE 7. Functional Logic Timing Diagram (Target Signal).

Functional logic relationship between primary clock input pulses at pin 14 and the output at pin 5.

Notes: Pin 1 shall be set at logic "0", pin 2 set at logic "1", pin 16 set to V_{DD}, Pin 8 set to V_{SS} (ground); logic levels on output pins 3, 4, 7 and 9 shall be as defined in Figures 5 and 6.

* Enable is an internal signal and not accessible at the package pins.

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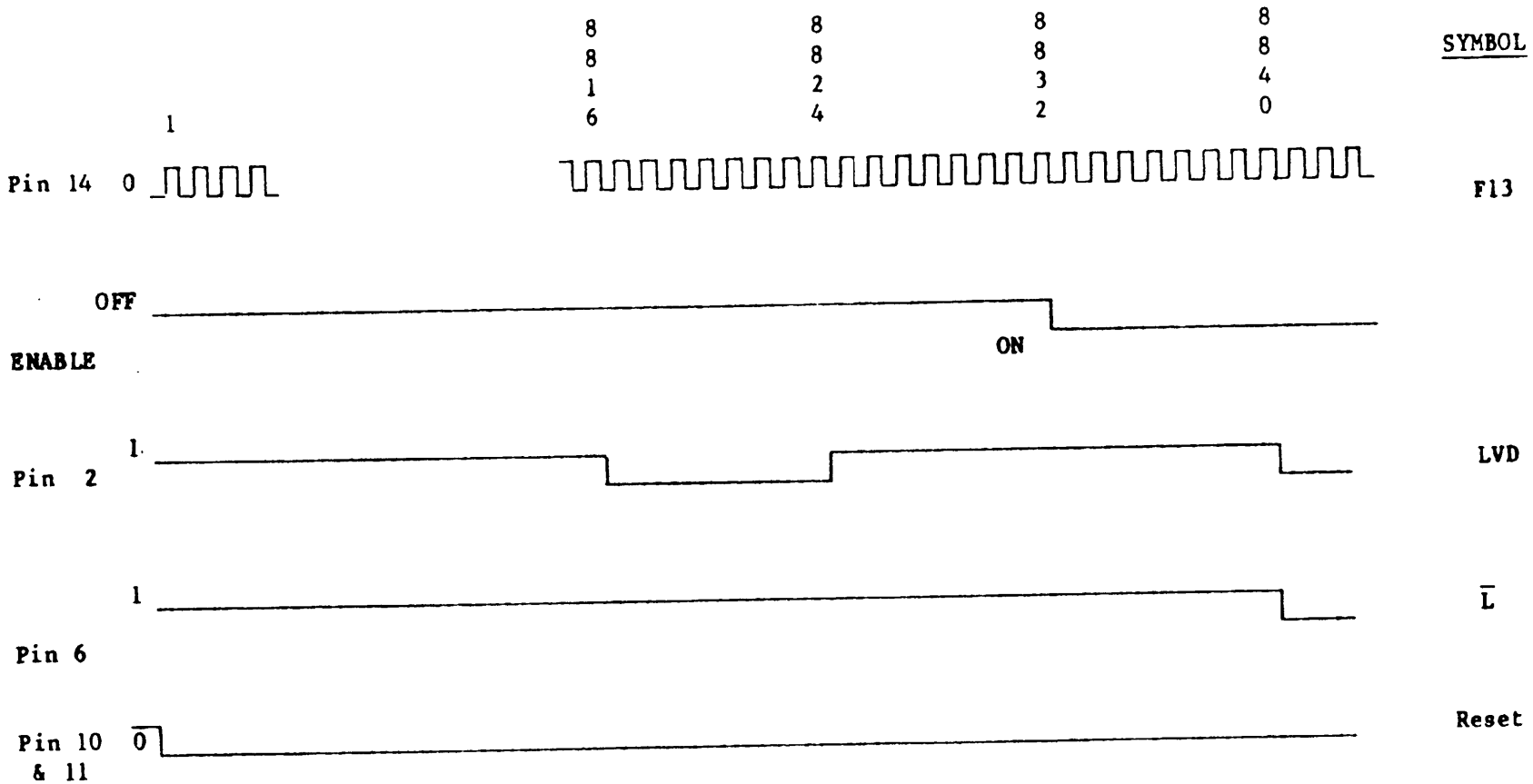


FIGURE 8. Functional Logic Timing Diagram (LVD).

Functional logic relationship between primary clock input pulses at pin 14 and ENABLE "ON" and between input pin 2 and output pin 6.

Notes: Pin 1 and 15 shall be set at logic "0", pin 16 set to V_{DD}, pin 8 set to V_{SS} (ground); logic levels on output pins 3, 4, 7 and 9 shall be as defined in Figures 5 and 6.

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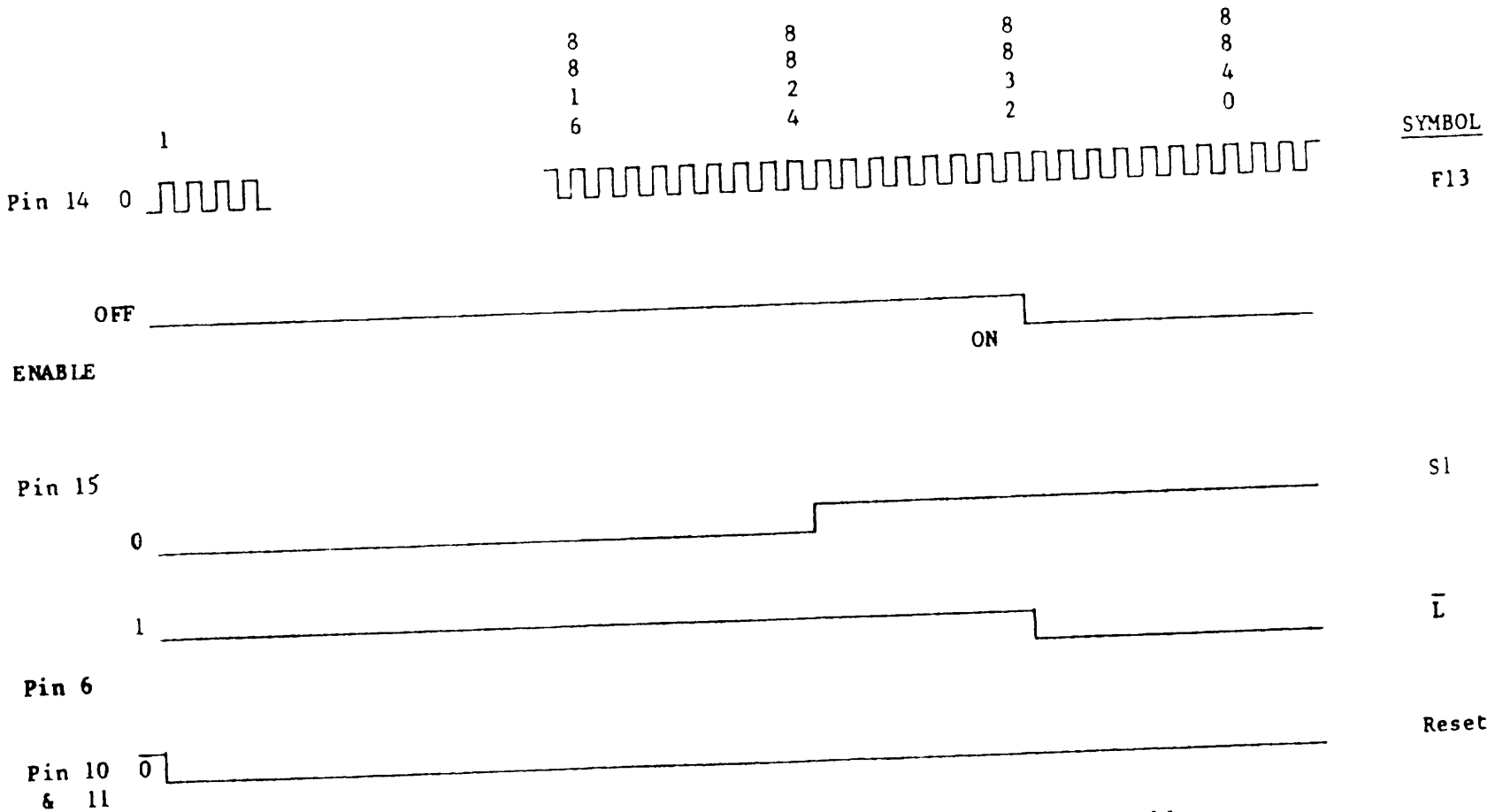


FIGURE 9. Functional Logic Timing Diagram - Target Signal, Pre-Enable.

Notes: Same as Figure 7 Notes.

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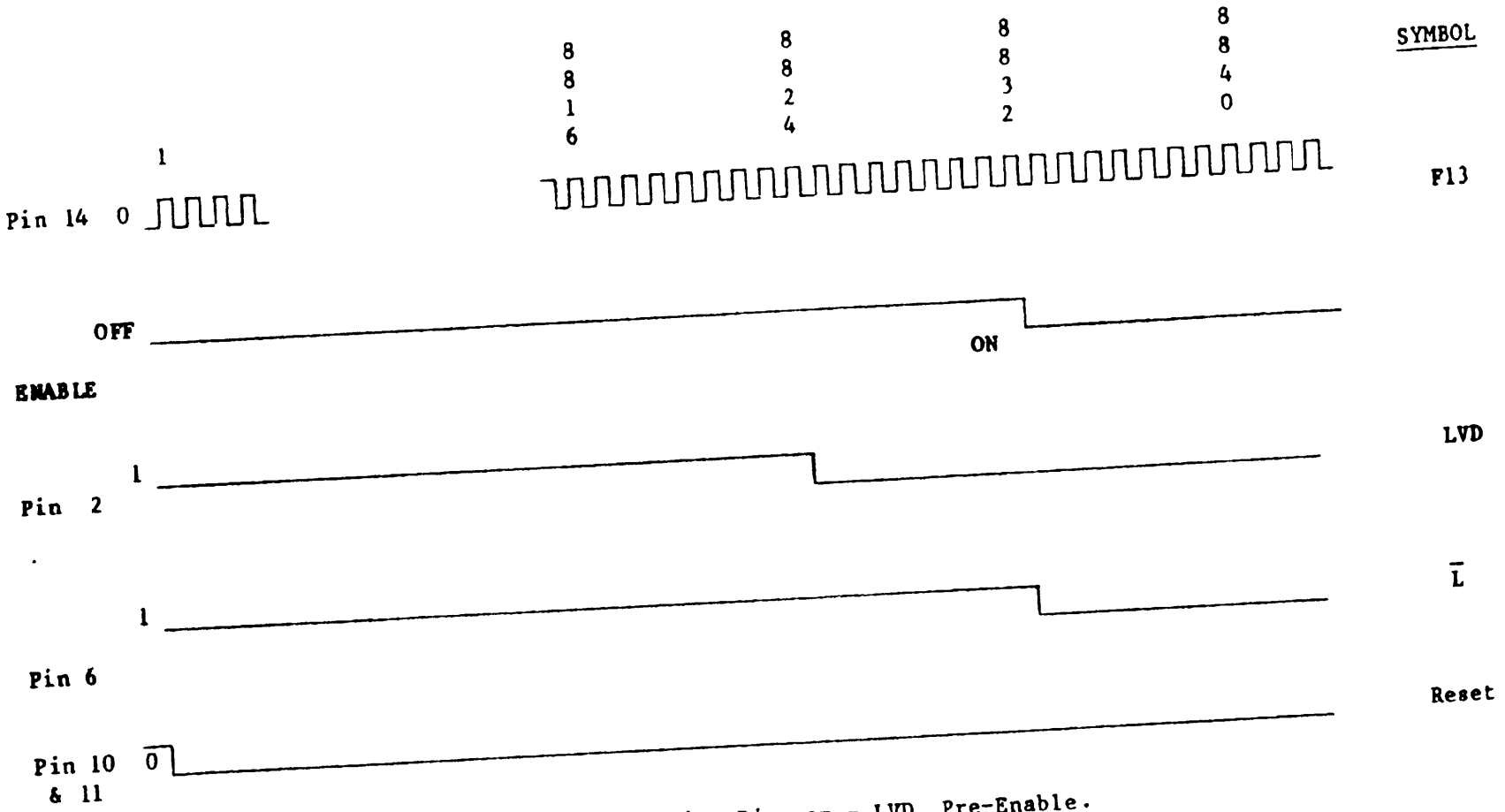


FIGURE 10. Functional Logic Timing Diagram - LVD, Pre-Enable.

Notes: Same as Figure 8 Notes.

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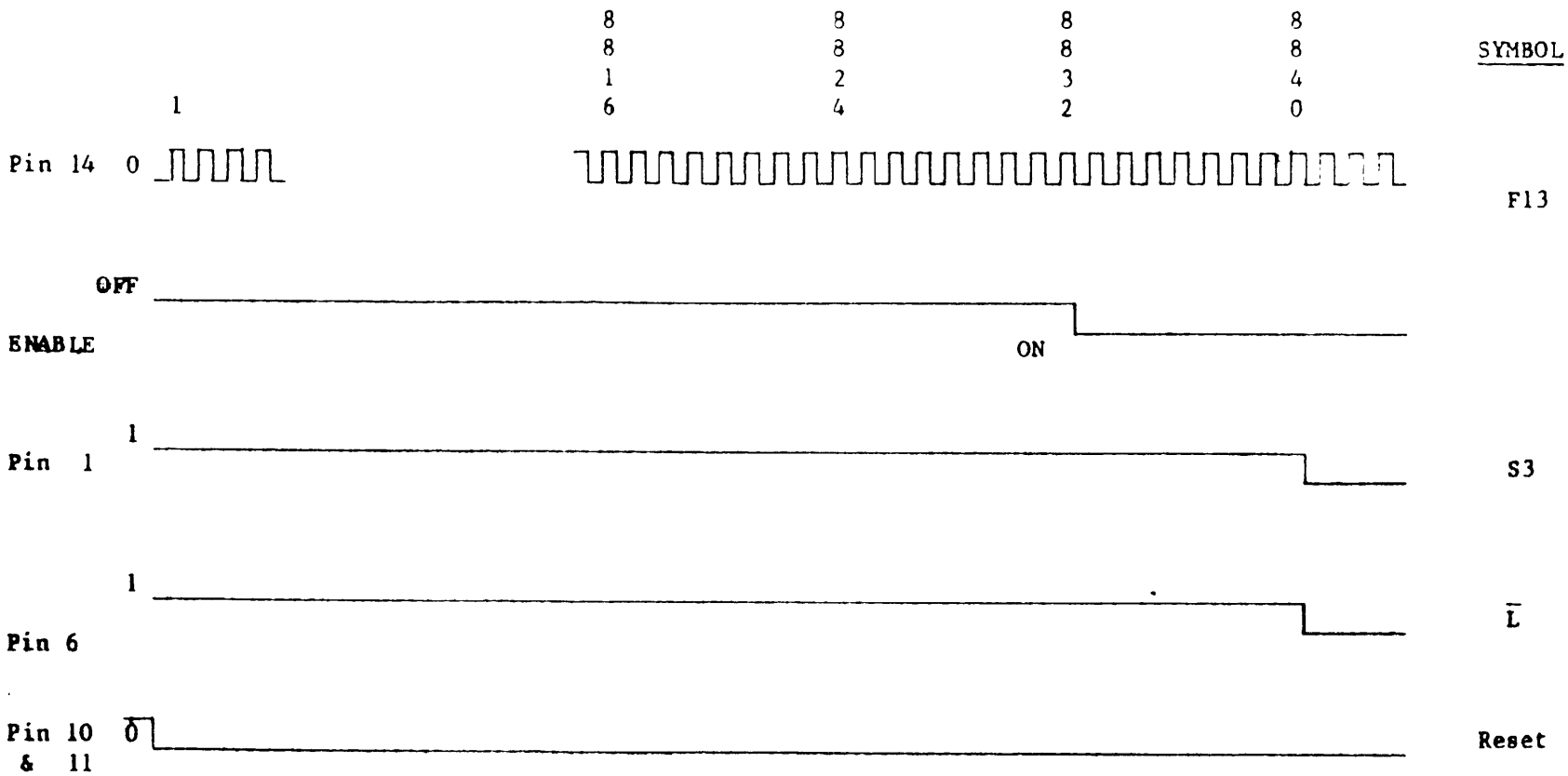


FIGURE 11. Functional Logic Timing Diagram (Antidisturbance, AD, Post Enable, High To Low).

Functional logic relationship between primary clock input pulses at pin 14 and and ENABLE "ON" and between input pin 1 and output pin 6.

Notes: Pins 2 shall be set at logic "1", pin 15 set at logic "0", pin 16 set to V_{DD}, pin 8 set to V_{SS} (ground); logic levels on output pins 3, 4, 7 and 9 shall be as defined in Figures 5 and 6.

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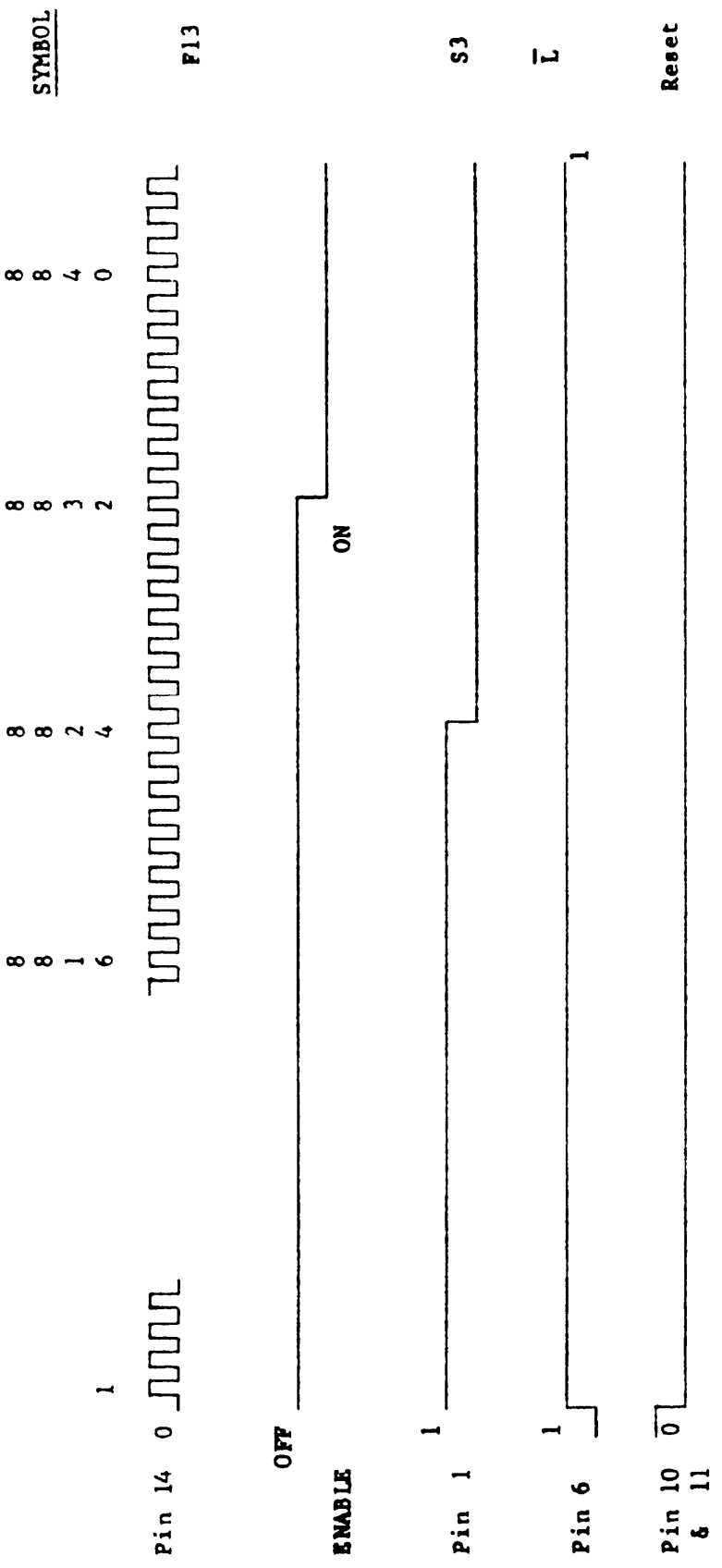


FIGURE 12. Functional Logic Timing Diagram (AD, Pre-Enable, High To Low).

Notes: Same as Figure 11 Notes.

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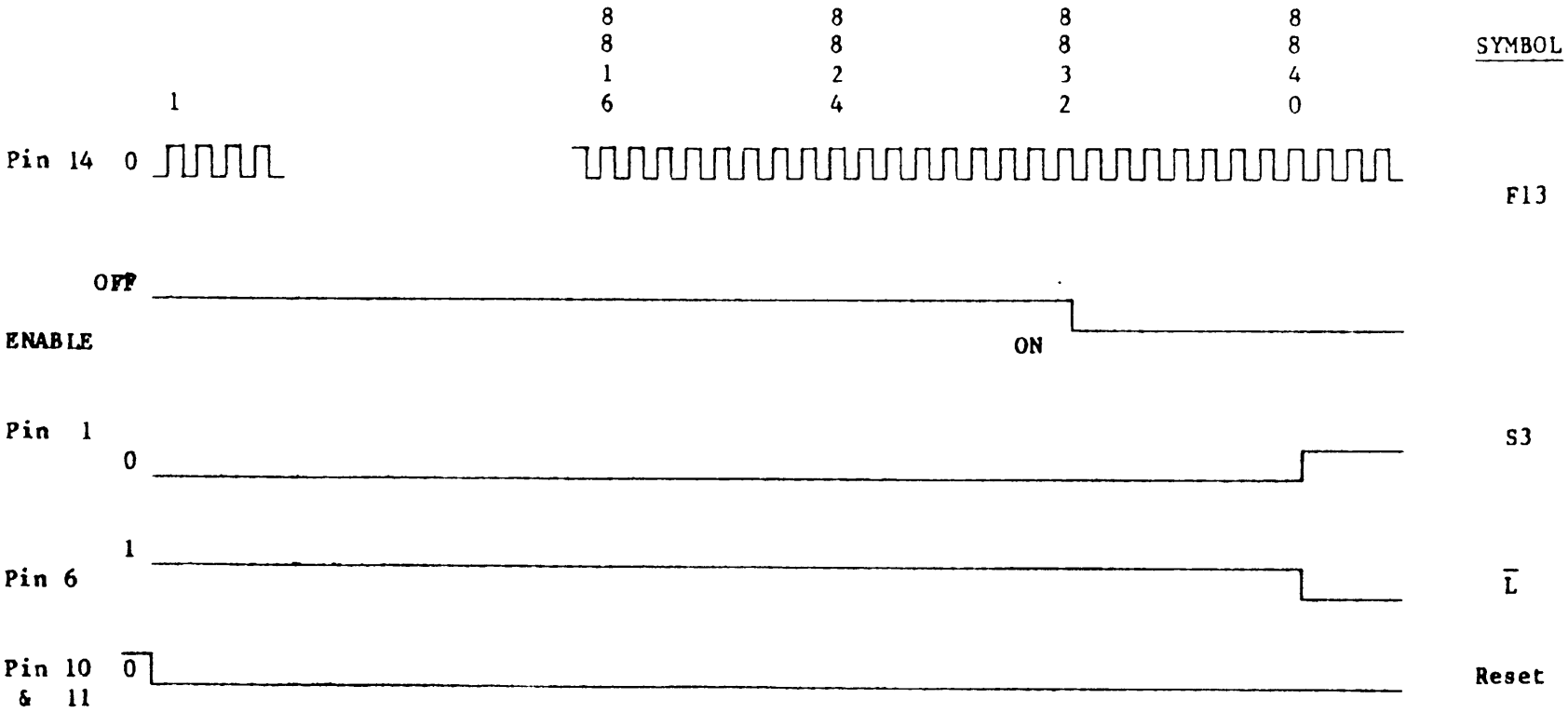


FIGURE 13. Functional Logic Timing Diagram (AD, Post Enable, Low to High).

Notes: Same as Figure 11 Notes.

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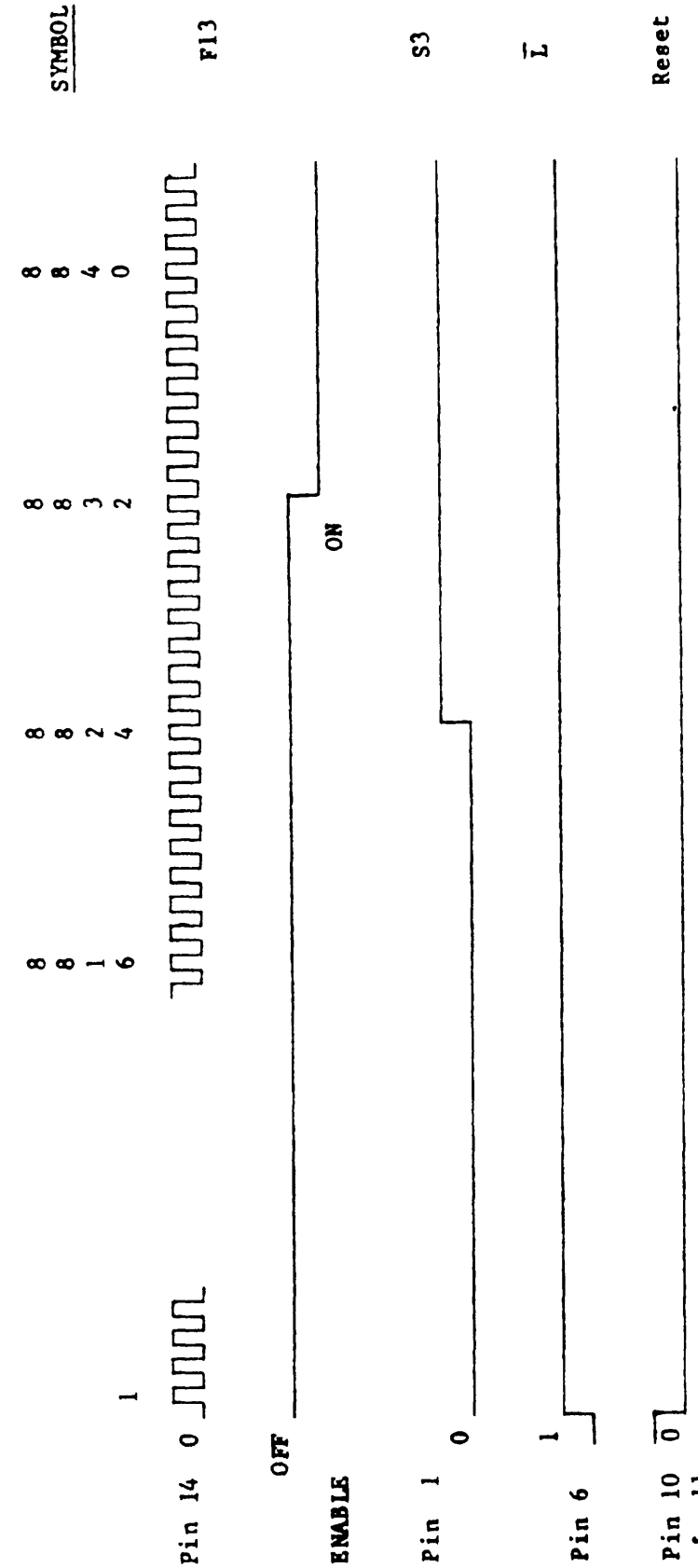


FIGURE 14. Functional Logic Timing Diagram (AD, Pre-Enable, Low to High).

Notes: Same as Figure 11 Notes.

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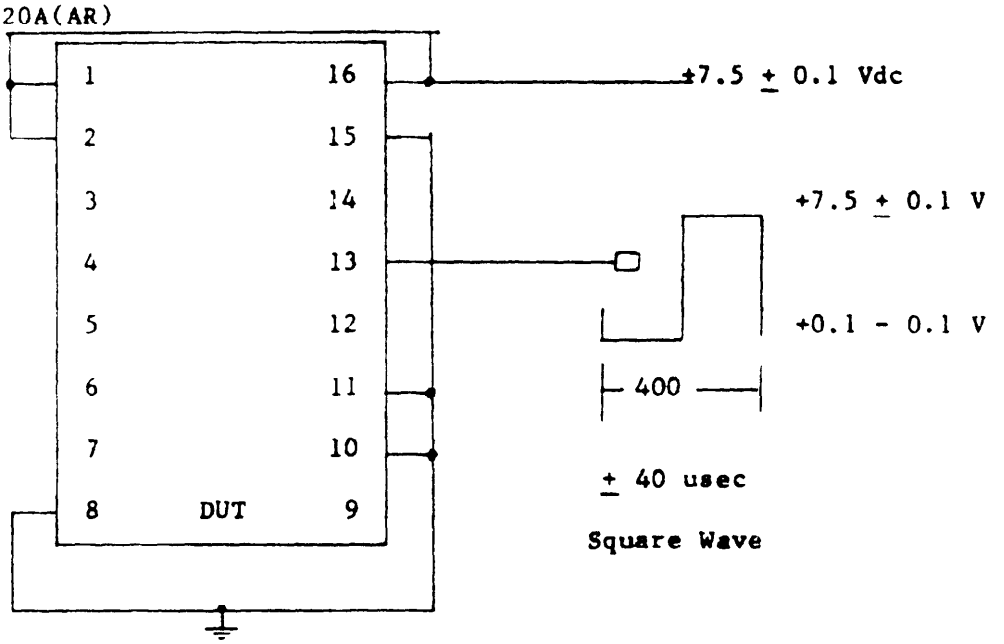
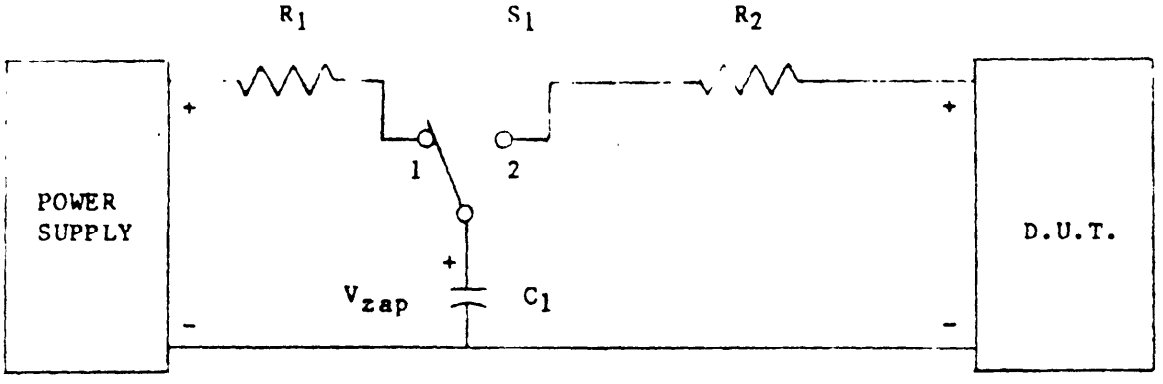


FIGURE 15. Burn-In and Steady State Life-Test Circuit.

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R_1 = 10 to 50 Meg ohms V_{zap} = 400 Volts charge on C_1
R_2 = 1.5 K-ohms S_1 = Hg. Wetted "bounceless" relay
C_1 = 100 pF

FIGURE 16. Static discharge test circuit

Static discharge (V_{zap}) test. Unless otherwise specified, one input terminal of the device under test (DUT) shall be subjected to a voltage pulse from a high voltage source. This destructive test shall be conducted as follows using the test circuit on Figure 16. All input terminals with static discharge protection shall be tested but only one input terminal per device shall be tested.

a. Measure input current, high level (I_{IH}) and low level (I_{IL}) 25°C ± 5°C at one input terminal of device under test (DUT). Also measure quiescent device current (I_L) at 25°C ± 5°C. The test limit for a single terminal measurement of I_{IH}, and I_{IL}, and I_L shall be increased a maximum of 10% of their value specified in the electrical characteristic Table I.

b. Apply the static discharge (V_{zap}) to the DUT in each of the following modes by charging C1 to V_{zap} with S1 in position 1 and then switching to position 2.

Modes	DUT +Terminal	DUT -Terminal
1	V_{DD}	Input
2	Input	V_{SS}

c. Within 24 hours repeat the I_L, I_{IL} and I_{IH} measurements on the same terminals as performed above. If the DUT exhibits leakage currents in excess of the specified limits at this time, it is defective. (Destructive test).

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TABLE VI. Signal Summary.

PACKAGE PIN NO.	INPUT/ OUTPUT	SIGNAL SYMBOL
1	IN	S_3
2	IN	LVD
3	OUT	Q
4	OUT	$\overline{C}$
5	OUT	F1
6	OUT	$\overline{L}$
7	OUT	T
8	SUPPLY	V_{SS}(GND)
9	OUT	PA
10	IN	R(RESET)
11	IN	R(RESET)
12	OUT	F11
13	IN	F12
14	OUT	F13
15	IN	S1
16	SUPPLY	V_{DD}(+)

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